Euclid's Elements — Book XI
Proposition 6

Formal Reconstruction — Technical Documentation

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If two straight lines be at right angles to the same plane, the straight lines will be parallel.

1 Introduction

Proposition XI.6 is the first proposition in the present Book XI development whose conclusion is a genuinely spatial parallelism statement. Two ambient lines l and m are perpendicular to the same plane π, at distinct feet B and D. Euclid concludes that the two lines are parallel.

In three-dimensional geometry this conclusion contains two logically different assertions. The lines must first be shown to be *coplanar*; only then can planar geometry prove that they do not meet. This distinction is essential in the formal reconstruction, because two disjoint ambient lines may be skew.

The current Lean proof follows the metric core of Euclid's construction closely:

$$\text{auxiliary perpendicular and equal segment} \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5}.$$

After XI.5 has supplied the missing common plane, the proof changes deliberately to an induced planar geometry and closes the disjointness argument by the neutral alternate-angle criterion.

2 Euclid's Statement

Euclid's Proposition XI.6 states:

> If two straight lines be at right angles to the same plane, the straight lines will be parallel.

In the classical configuration the two perpendiculars are AB and CD, meeting the reference plane at B and D. The formal theorem makes the nondegenerate case explicit by assuming

$$B \neq D.$$

3 Classical Configuration

Join the feet B and D. In the reference plane construct through D a line DE perpendicular to BD, and choose E so that

$$DE \cong AB.$$

Join BE, AE, and AD.

Because AB is perpendicular to the reference plane, it is perpendicular to every line of that plane through B, in particular to BD and BE. Hence

$$\angle ABD, \qquad \angle ABE$$

are right angles.

Likewise CD is perpendicular to the reference plane at D, so it is perpendicular to both DB and DE.

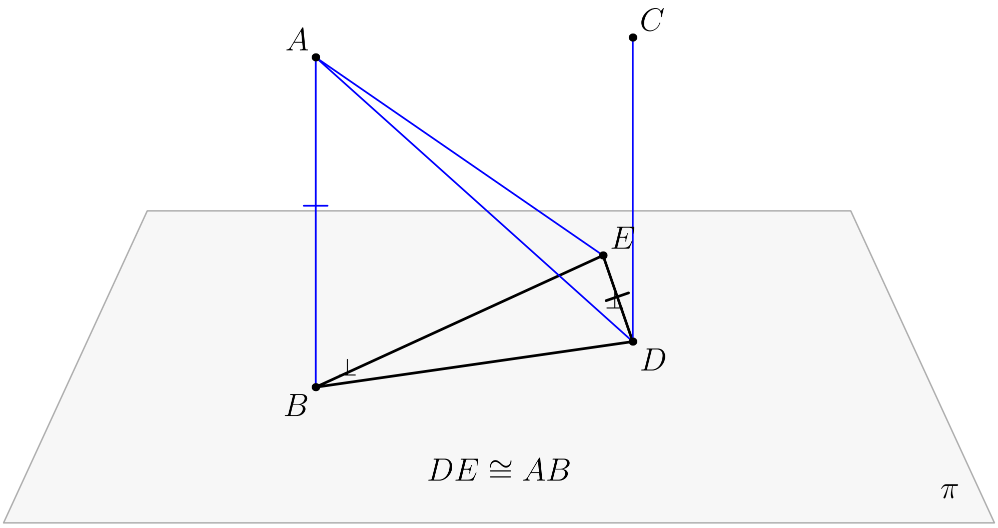

Figure 1: The classical configuration of Proposition XI.6. The black geometry lies in the reference plane π: B, D, E, the transversal BD, and the auxiliary perpendicular DE. The blue lines and connectors leave π. The construction $DE \cong AB$ prepares the SAS comparison of BAD and DEB; the resulting equality $AD \cong EB$ prepares the SSS comparison of DEA and BAE.

4 Classical Proof

The classical metric part consists of two triangle-congruence steps.

First,

$$AB \cong DE, \qquad BD \cong DB,$$

and the included angles

$$\angle ABD, \qquad \angle EDB$$

are both right. Proposition I.4 therefore gives

$$AD \cong EB.$$

Second, the triangles DEA and BAE have

$$DE \cong BA, \qquad EA \cong AE, \qquad DA \cong BE.$$

Proposition I.8 gives

$$\angle EDA \cong \angle ABE.$$

Since $\angle ABE$ is right, $\angle EDA$ is right; hence

$$ED \perp DA.$$

Now at the point D the line ED is perpendicular to the three lines

$$DB, \qquad DA, \qquad DC.$$

Proposition XI.5 therefore places those three lines in one plane. Proposition XI.2 identifies the triangle plane determined by the intersecting lines, so AB is brought into the same plane as BD and DC. Finally

$$\angle ABD = \angle BDC = 90^\circ,$$

and Proposition I.28 yields

$$AB \parallel CD.$$

A compact classical dependency graph is therefore

$$\begin{gathered}
\text{construct } DE \perp BD, \quad DE \cong AB \\
\Downarrow \\
\text{I.4} \\
\Downarrow \\
AD \cong EB \\
\Downarrow \\
\text{I.8} \\
\Downarrow \\
ED \perp DA \\
\Downarrow \\
\text{XI.5} \\
\Downarrow \\
DB, DA, DC \text{ coplanar} \\
\Downarrow \\
\text{XI.2} \\
\Downarrow \\
AB, BD, DC \text{ coplanar} \\
\Downarrow \\
\text{I.28} \\
\Downarrow \\
AB \parallel CD.
\end{gathered}$$

The construction of the auxiliary perpendicular and equal segment is part of the earlier Book I construction machinery. The substantive local congruence steps are I.4 and I.8.

5 What is genuinely three-dimensional here?

The difficult point is not the planar theorem that two perpendiculars to one transversal are parallel. That statement cannot even be applied until one knows that the two candidate parallel lines lie in one plane.

In ambient three-space the information

$$l \perp BD, \qquad m \perp BD$$

does not by itself constitute a parallelism proof: without a coplanarity theorem the lines could still be skew.

The formal proof therefore separates XI.6 into

$$\boxed{\text{coplanarity}} \qquad + \qquad \boxed{\text{planar disjointness}}.$$

The first box is the genuinely spatial content and is exactly where Proposition XI.5 is used.

6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_6
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (pi : S.Plane)
    (l m : Geo.Line)
    (B D : Geo.Point)
    (hBD : Ne B D)
    (hLperp :
      HilbertLinePerpendicularPlaneAt
        Geo l pi B)
    (hMperp :
      HilbertLinePerpendicularPlaneAt
        Geo m pi D) :
    HilbertSpaceLinesParallel Geo l m
```

The target relation is part of the reusable spatial interface `Hilbert3DInterface`:

```
def HilbertSpaceLinesParallel
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m
```

Thus “parallel in space” means exactly:

1. there is a plane containing both lines;
2. the two lines are disjoint.

The first clause is not redundant. If spatial parallelism were defined only by ambient disjointness, skew lines would incorrectly satisfy the definition.

7 Spatial / Planar Architecture Used

The proof uses three layers which must remain distinct.

7.1 Ambient three-space

The ambient objects and relations include

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

In particular, segment congruence between configurations carried by different planes is handled through `HilbertSpaceCongruence`.

7.2 Induced plane geometry

Whenever a planar theorem is needed, the relevant ambient plane σ or π is converted into

```
PlanePoint
PlaneLine
PlaneGeo
```

and ordinary planar incidence, order, right-angle, side-of-line, and triangle theorems are used there.

There is deliberately no global ambient instance

```
HilbertOrder Geo
HilbertCongruence Geo
```

for the three-dimensional space.

7.3 Plane–space bridges

The proof repeatedly uses bridges such as

```
planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
```

```
planeGeo_primCollinear_to_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_sameRay_iff_ambient
```

These are not cosmetic casts. They state which geometric fact is being used in which plane slice.

8 Proof Architecture of the Reconstruction

8.1 Step 1: Euclid's auxiliary construction in the reference plane

Choose a point $A \neq B$ on the first normal l. Inside the reference plane π, construct the line

$$d = BD.$$

Through D construct a line e perpendicular to d, and lay off on e a point E such that

$$DE \cong AB.$$

This is packaged by

```
hilbert_XI6_auxiliary_perpendicular_equal_segment
```

The line d and the auxiliary line e are constructed in `PlaneGeo pi`. The segment copied onto DE, however, may have the spatial endpoint A; that copy is therefore performed by `HilbertSpaceCongruence.segment_construction`.

8.2 Step 2: the first spatial SAS

Since $l \perp \pi$ at B and $d \subset \pi$ passes through B,

$$l \perp d.$$

The auxiliary construction gives

$$e \perp d$$

at D.

The helper

```
hilbert_XI6_space_linesPerpendicularAt_right_angle_of_points
```

normalizes these line–line perpendicularities to the concrete right angles

$$\angle ABD, \qquad \angle EDB.$$

The spatial right-angle comparison then supplies

$$\angle ABD \cong \angle EDB.$$

With
$$BA \cong DE, \qquad BD \cong DB,$$
the spatial SAS theorem gives
$$AD \cong EB.$$

This whole block is packaged by

```
hilbert_XI6_first_SAS
```

and uses

```
hilbert_space_sas_third_side_and_angle
```

for the actual triangle comparison.

8.3 Step 3: the second spatial SSS and the new right angle

The next comparison is between the triangles DEA and BAE. The three side relations are

$$DE \cong BA, \qquad EA \cong AE, \qquad DA \cong BE.$$

The spatial SSS theorem gives
$$\angle EDA \cong \angle ABE.$$
Because $AB \perp \pi$ and $BE \subset \pi$, the angle ABE is right. The rightness is transported to EDA, yielding
$$ED \perp DA.$$

This block is packaged by

```
hilbert_XI6_second_SSS_right_angle
```

The proof must create explicit carrier planes for the relevant triangles. The right-angle transport is handled by

```
hilbert_XI6_space_right_angle_transport_in_plane
```

rather than by installing a forbidden ambient `HilbertCongruence Geo` instance.

8.4 Step 4: Proposition XI.5 supplies coplanarity

At D, the line $e = DE$ is now perpendicular to

$$d = DB, \qquad a = DA, \qquad m.$$

The last perpendicularity follows because $m \perp \pi$ and $e \subset \pi$.

Proposition XI.5 is applied exactly here:

```
euclid_proposition_11_5
```

It produces a plane σ containing

$$d, \qquad a, \qquad m.$$

Since $A \in a$ and $B \in d$, both A and B lie in σ. The line $l = AB$ therefore also lies in σ.

The result is isolated as

```
hilbert_XI6_normals_to_same_plane_coplanar
```

with conclusion

```
exists sigma : S.Plane,
  HilbertLineInPlane Geo l sigma /\
  HilbertLineInPlane Geo m sigma
```

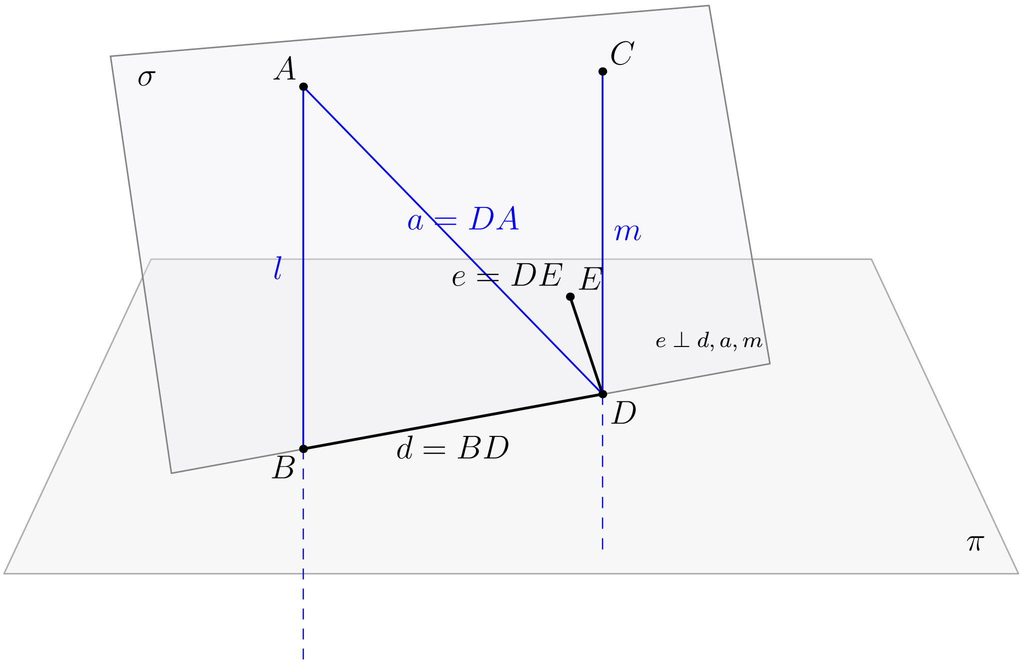

Figure 2: The genuinely spatial step in the formal XI.6 proof. The black lines $d = BD$ and $e = DE$ lie in the reference plane π. The blue lines $a = DA$, $m = DC$, and $l = AB$ leave π. At D, the line e is perpendicular to d, a, m, so XI.5 produces the second plane σ containing d, a, m. Because σ contains A and B, it also contains l. The lines l and m are therefore coplanar.

8.5 Step 5: the remaining problem is planar

After Step 4, both l and m lie in σ. The common transversal $d = BD$ lies in σ as well, and

$$l \perp d \text{ at } B, \qquad m \perp d \text{ at } D.$$

The rest of the proof is carried out entirely inside

`PlaneGeo Geo sigma`.

The helper

```
hilbert_XI6_coplanar_normals_disjoint
```

chooses nonfoot points on l and m, establishes the required side-of-line orientation relative to d, compares the two right angles, and applies

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

to obtain planar parallelism.

The final conversion proves that the actual ambient carriers l and m are disjoint.

8.6 Step 6: assemble spatial parallelism

The public theorem now has exactly the two pieces required by `HilbertSpaceLinesParallel`:

$$l, m \subset \sigma$$

and

$$\mathrm{HilbertLinesDisjoint}(l, m).$$

Therefore

$$\mathrm{HilbertSpaceLinesParallel}(l, m).$$

9 Lean Formalization

The final theorem body is short because the geometric work has been isolated into two major helpers. First,

```
hilbert_XI6_normals_to_same_plane_coplanar
```

returns a plane `sigma` containing both ambient normals. Then

```
hilbert_XI6_coplanar_normals_disjoint
```

proves that their ambient carriers are disjoint. The public theorem packages exactly these three fields of `HilbertSpaceLinesParallel`: containment of l, containment of m, and disjointness.

The production module imports

```
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition27
```

The first import supplies the decisive spatial coplanarity theorem. The second supplies the neutral alternate-angle machinery used in the final induced plane.

10 Hidden Formal Data

The classical diagram suppresses several obligations which the Lean proof must make explicit.

- The feet B, D are distinct.
- The point A selected on l is outside the reference plane π.
- The auxiliary point E is nondegenerate and lies both on e and in π.
- Every triangle used in SAS or SSS carries an explicit noncollinearity proof.
- Spatial triangle comparisons require an explicit target carrier plane.
- Right-angle transport between ambient configurations is performed through an explicit `PlaneGeo`, not through an ambient planar congruence instance.
- Before XI.5 can be applied, the three lines d, a, m must be proved distinct in the required places and must all pass through D.
- The plane σ returned by XI.5 must be shown to contain both A and B, so that the whole carrier $l = AB$ lies in σ.
- In the final planar proof, the chosen points on l and m must be oriented on opposite sides of the transversal d. The proof performs an explicit SameSide/OppositeSide case split.
- Point-pair parallelism inside `PlaneGeo sigma` must finally be converted into disjointness of the original ambient carriers.

These obligations are geometric. They are the incidence, order, and plane-membership data that a conventional three-dimensional drawing usually leaves implicit.

11 Relationship with Earlier Book XI Propositions

Proposition XI.5 is a direct formal dependency:

$$\boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

XI.5 itself depends on XI.4, and XI.6 also reuses several XI.4 normalization helpers for perpendicular lines and right angles. Consequently the effective Book XI chain is

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

The classical proof also invokes XI.2 when it moves from the three coplanar lines through D to the plane containing the triangle ABD. The Lean proof does not need a separate formal call to Proposition XI.2: the spatial incidence API directly constructs and tracks the relevant carrier plane.

12 Relationship with Book I / Book Zero / Hilbert Infrastructure

The metric part of the formal proof reproduces the roles of Euclid I.4 and I.8 through the spatial SAS and SSS infrastructure:

```
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

Right angles and their transport are handled synthetically; no angle measure is introduced.

The classical final step is I.28. The production Lean file instead imports `Proposition27` and uses the established neutral alternate-angle theorem

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

inside the explicitly constructed plane σ. Thus the formal module dependency and the classical Euclid proposition-number dependency are not identical, even though they close the same planar configuration.

Book Zero enters only transitively through the established Hilbert geometric infrastructure. No new Book Zero axiom or theorem is added by XI.6.

13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_6

#print axioms Geometry.euclid_proposition_11_6
```

and Lean reports

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

This is the same reported axiom profile as Propositions XI.4 and XI.5.

Therefore:

- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;
- Euclidean parallel axiom / Hilbert Group IV: not used;

- new proposition-local helper theorems: yes;
- new `Hilbert3DAxioms` declaration: no;
- reusable `Hilbert3DInterface` declaration: `HilbertSpaceLinesParallel`, promoted from the XI.6 proposition module for reuse by XI.7 and later spatial results;
- new `PlaneGeo` bridge theorem: no;
- `sorry`/`admit`: no.

The inherited `Geometry.bookZero_nullSegment1` dependency is not specific to XI.6.

The production theorem compiled successfully, its axiom audit was run successfully, and the full repository `lake build` was reported clean at this checkpoint.

14 Dependency Summary

The classical local chain is

$$\begin{array}{c}\text{Book I perpendicular/equal-segment construction}\\ \downarrow \\ \text{I.4 (SAS)} \\ \downarrow \\ \text{I.8 (SSS)} \\ \downarrow \\ \text{XI.5} \\ \downarrow \\ \text{XI.2} \\ \downarrow \\ \text{I.28} \\ \downarrow \\ \text{XI.6.}\end{array}$$

The actual formal chain is

$$\begin{gathered}
\text{Hilbert incidence / order / congruence infrastructure}\\
\downarrow\\
\text{spatial primitive / incidence / order / congruence}\\
\downarrow\\
\text{PlaneGeo slices and plane–space bridges}\\
\downarrow\\
\text{XI.4 right-angle and perpendicular normalizers}\\
\downarrow\\
\text{auxiliary } d = BD,\ e \perp d,\ DE \cong AB\\
\downarrow\\
\text{spatial SAS: } AD \cong EB\\
\downarrow\\
\text{spatial SSS: } ED \perp DA\\
\downarrow\\
\texttt{euclid_proposition_11_5}\\
\downarrow\\
l, m \text{ coplanar in } \sigma\\
\downarrow\\
\text{PlaneGeo}(\sigma) \text{ side/order + alternate angles}\\
\downarrow\\
\text{ambient carrier disjointness}\\
\downarrow\\
\texttt{HilbertSpaceLinesParallel}\\
\downarrow\\
\texttt{euclid_proposition_11_6}.
\end{gathered}$$

15 Conclusion

Proposition XI.6 is a useful boundary theorem for the three-dimensional architecture. Its metric core is recognizably Euclidean: construct an auxiliary perpendicular and an equal segment, use SAS, then SSS, and obtain a third perpendicular at D.

The genuinely spatial step is XI.5. It converts three perpendicularities through one point into the common plane in which the final parallelism argument is allowed to take place. Lean makes this logical boundary explicit rather than allowing the diagram to hide it.

The final definition of spatial parallelism records the same distinction: coplanarity is proved first and disjointness second. No coordinates, vectors, scalar products, numerical angle measure, continuity principle, new spatial axiom, or Euclidean parallel axiom is introduced.